

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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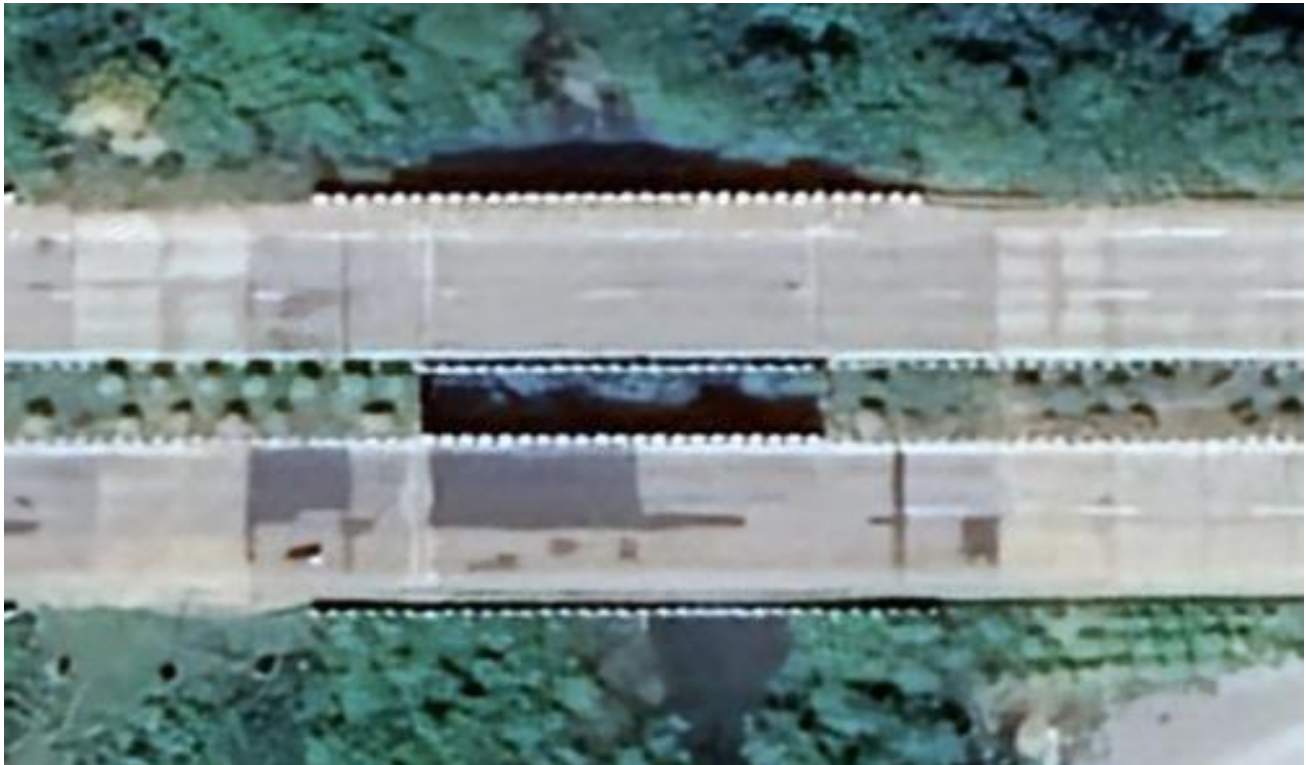
3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 917+700

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

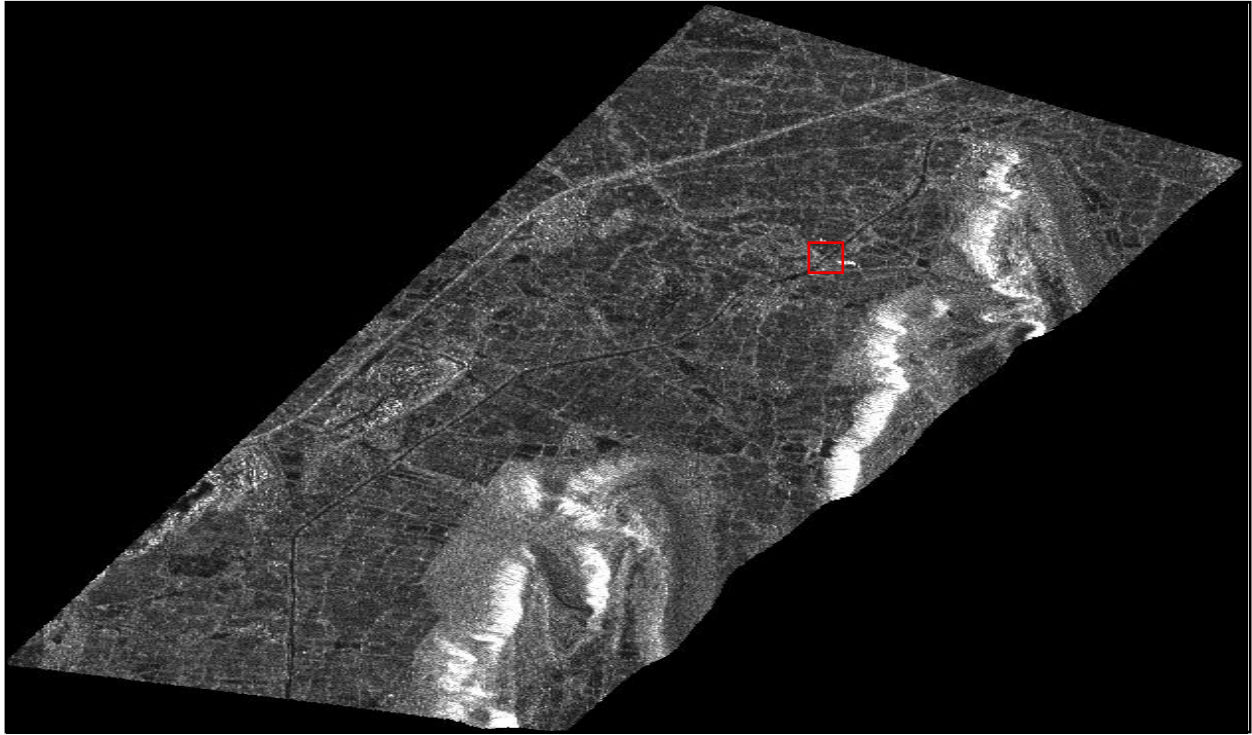
Flight Direction - DESCENDING

Polarization – VV+VH

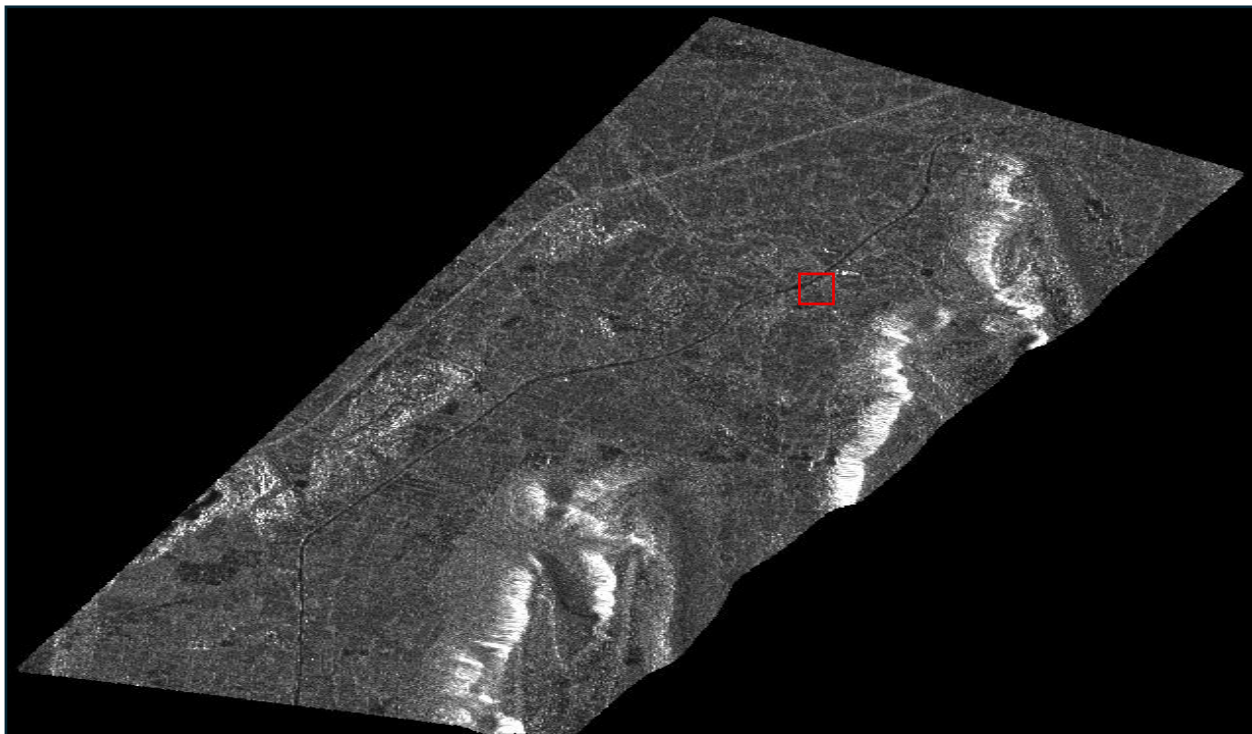
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

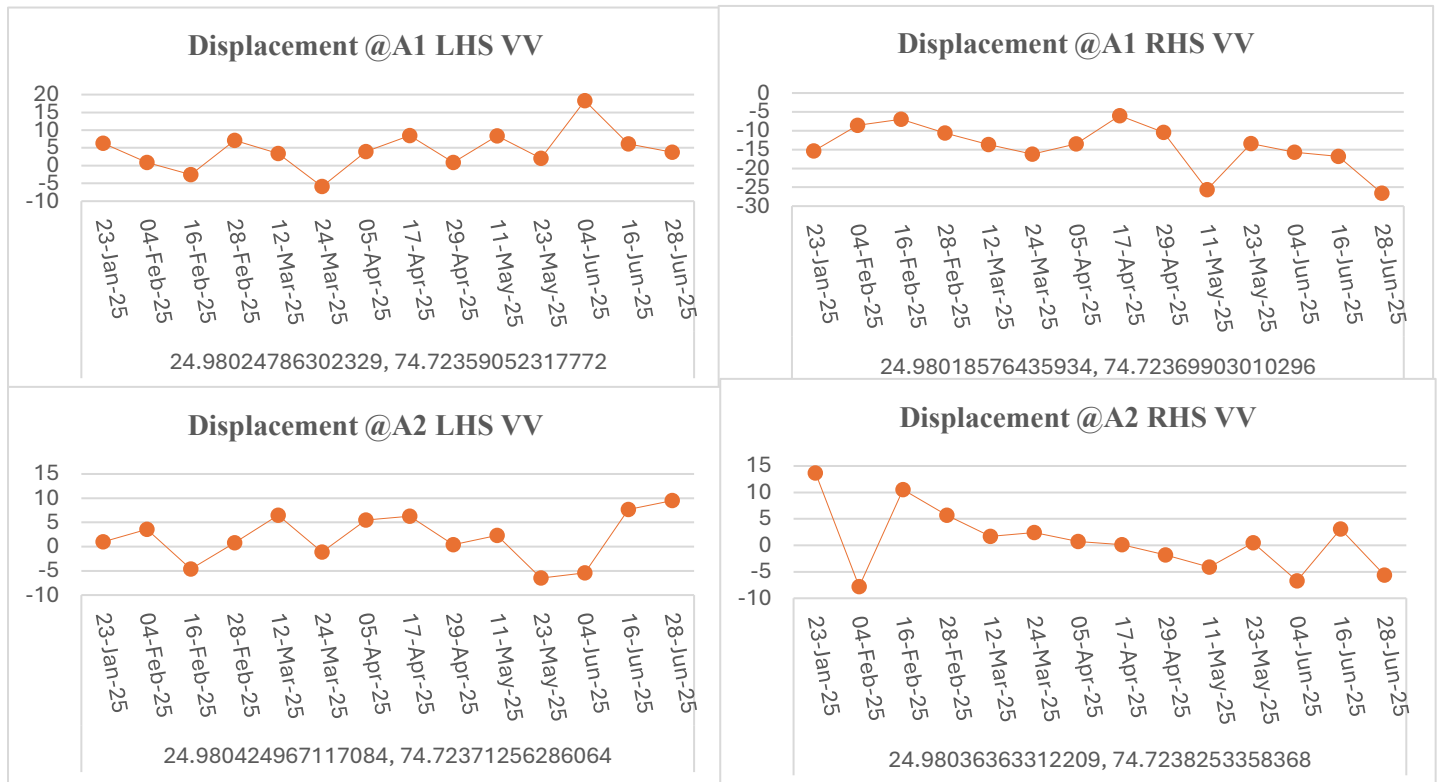
MNB – CH-917+700

Length – 21.9M

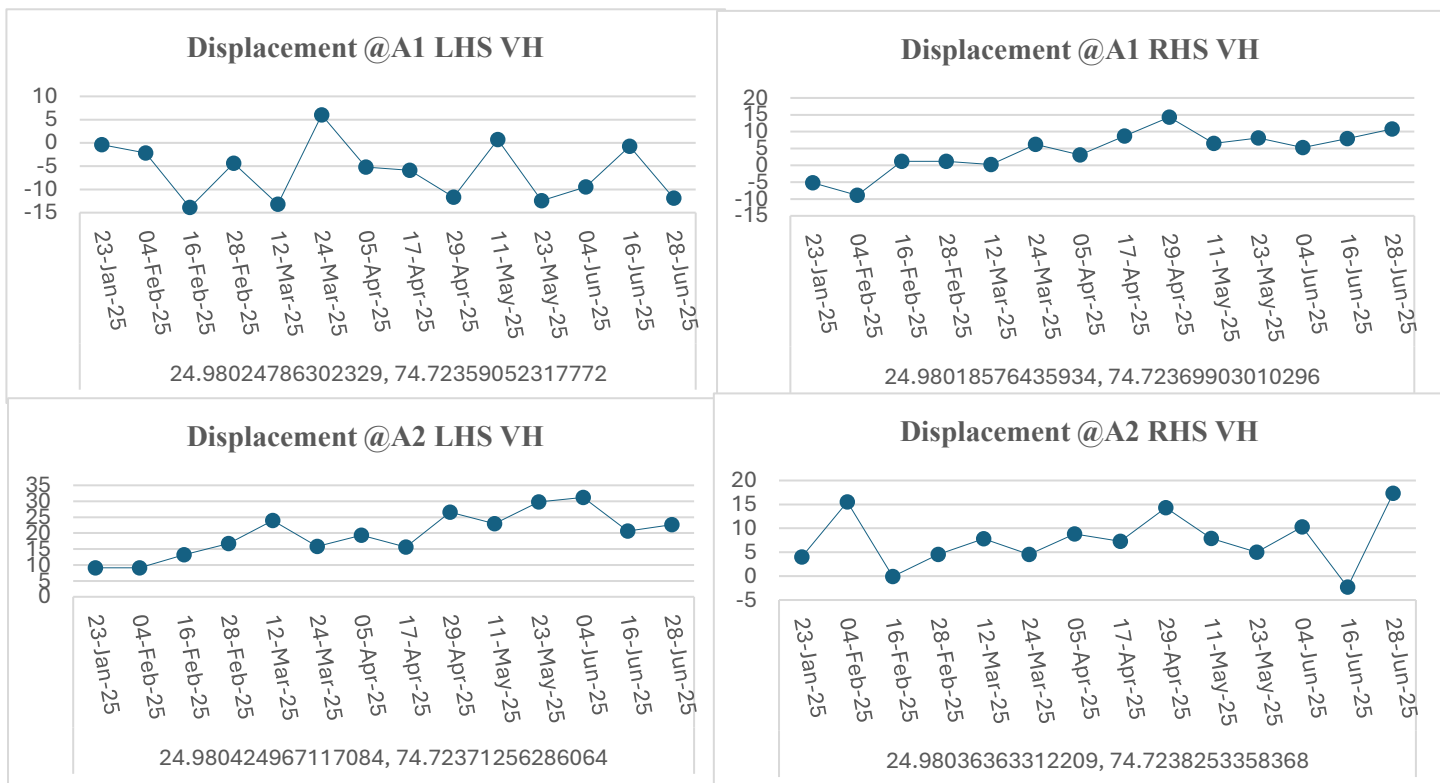


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

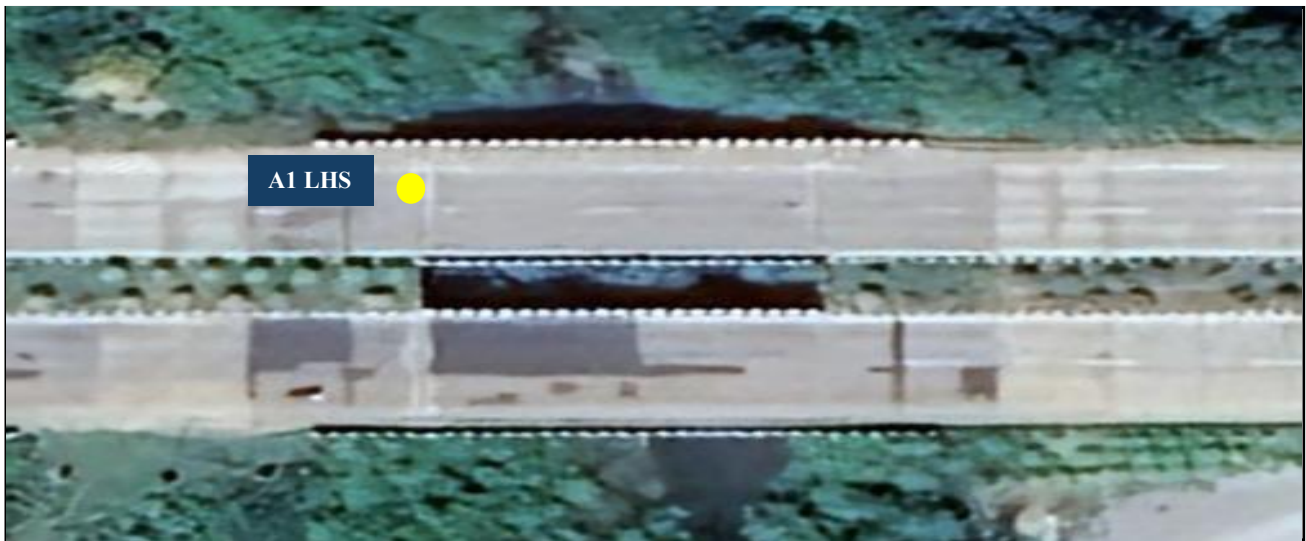
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



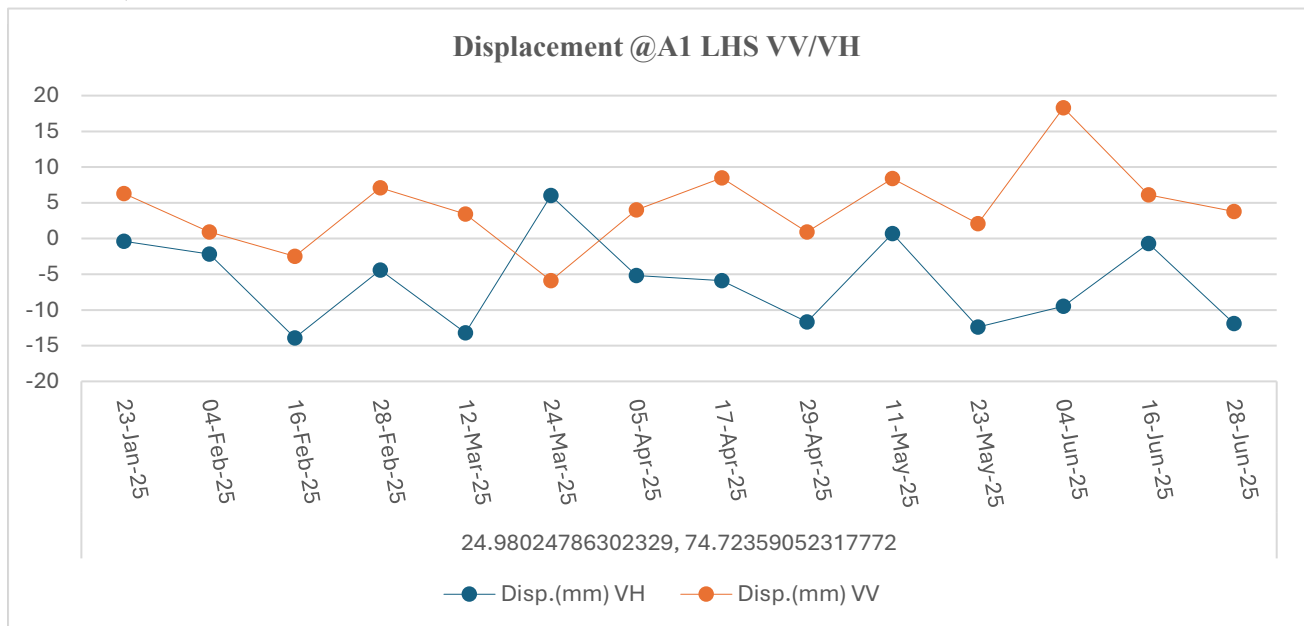
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.98024786302329, 74.72359052317772)



Key Observations:

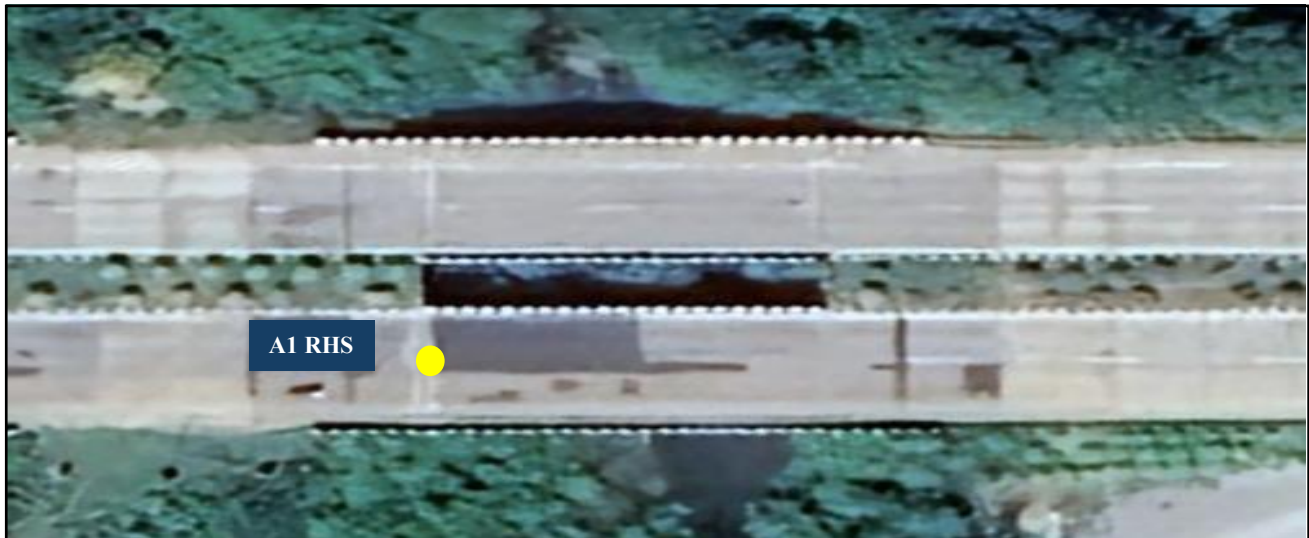
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with a predominantly positive tendency over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with strong negative dominance, ranging roughly between +6 mm and –15 mm.

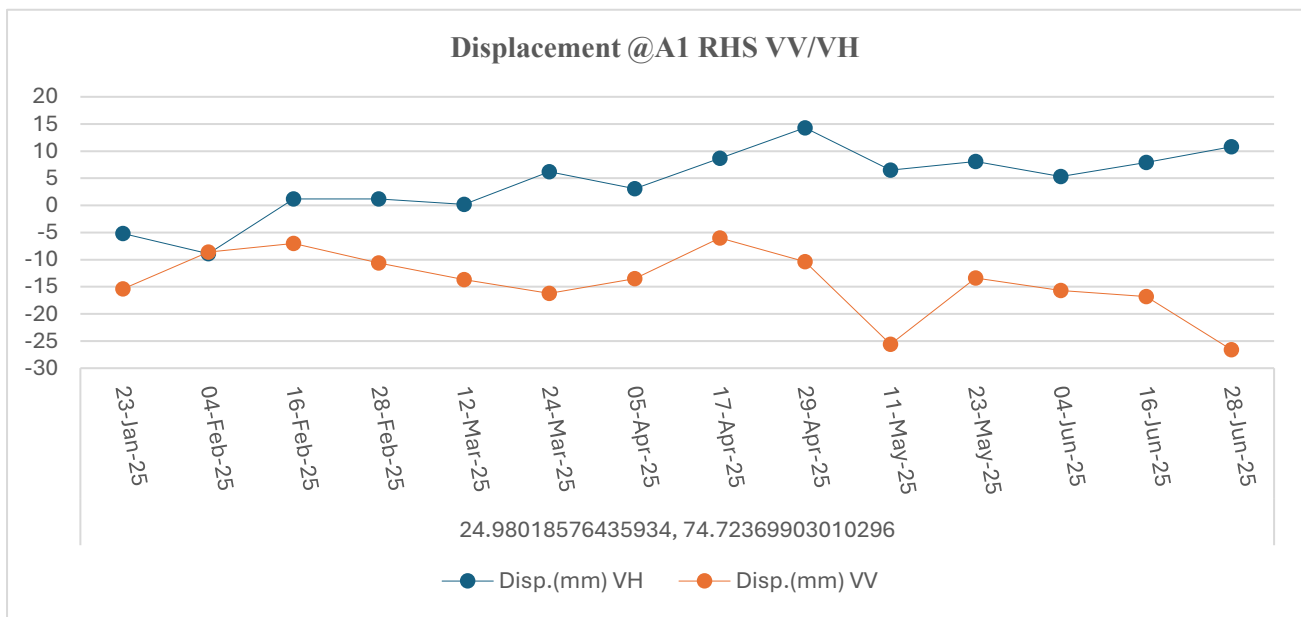
Critical Events & Anomalies:

- **March 2025:** VH again deteriorates to –14 mm, while VV remains slightly positive (+3 mm), highlighting a lateral-dominant deformation phase.
- **April 2025:** VH declines again to –6 mm, while VV rises to +9 mm, indicating renewed uplift under continued lateral instability.
- **June 2025:** VH drops again to –12 mm, while VV stabilizes near +5 mm, confirming continued lateral settlement toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.98018576435934, 74.72369903010296)



Key Observations:

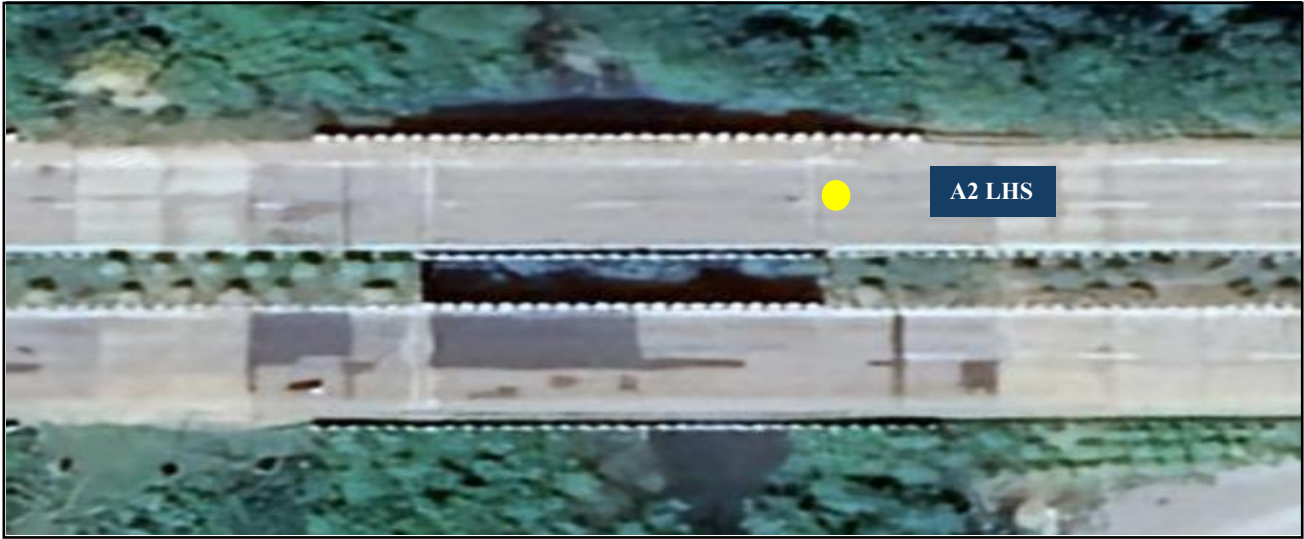
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability with a clear positive dominance, ranging roughly between -5 mm and $+14$ mm.
- **Vertical displacement (VV)** Vertical displacement is consistently negative throughout the monitoring period, indicating persistent vertical settlement at the A1 RHS location.

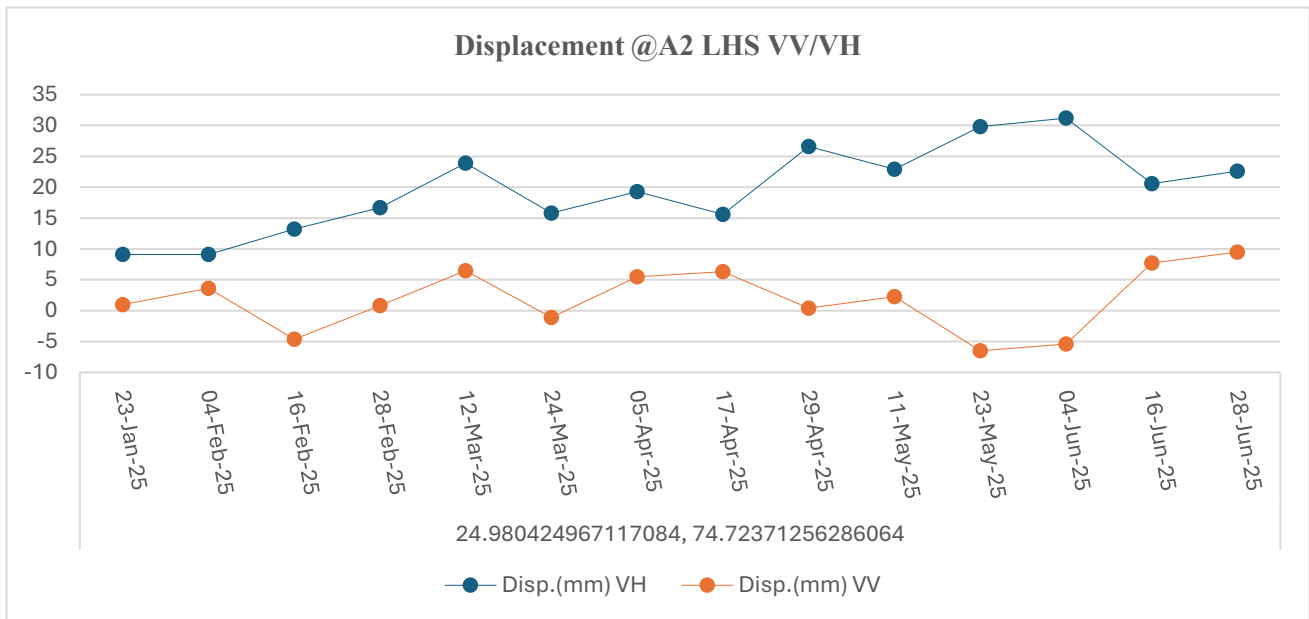
Critical Events & Anomalies:

- **January 2025:** VH begins slightly negative (-5 mm), while VV is already strongly negative (-16 mm), indicating early coupled lateral and vertical settlement.
- **May 2025:** VH fluctuates between $+5$ and $+7$ mm, while VV remains deeply negative (-16 to -18 mm), showing continued settlement dominance.
- **June 2025:** VH increases again to $+11$ mm, while VV drops back toward -26 mm, confirming persistent vertical settlement combined with ongoing lateral displacement toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.980424967117084, 74.72371256286064)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with a strong and sustained positive dominance, ranging roughly between +10 mm and +31 mm.
- **Vertical Displacement (VV)** Vertical displacement shows low to moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –8 mm to +10 mm.

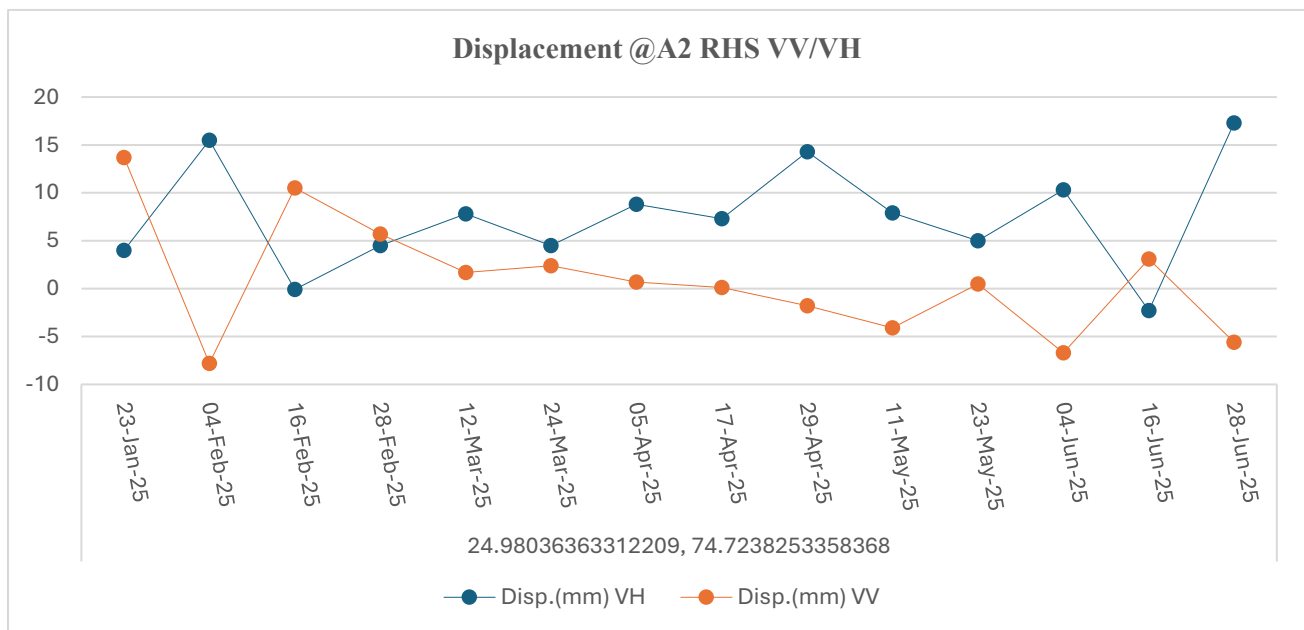
Critical Events & Anomalies:

- **April 2025:** VH reaches +30–31 mm, the maximum observed horizontal displacement, while VV drops sharply to –8 mm, representing a highly asymmetric deformation event dominated by lateral movement.
- **June 2025:** VH stabilizes around +23 mm, while VV remains positive (+10 mm), confirming persistent lateral displacement with non-degrading vertical response toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.98036363312209, 74.7238253358368)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** exhibits moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –8 mm to +14 mm.
- **Horizontal (VH) displacement** Horizontal displacement shows high variability with an overall positive dominance, ranging roughly between 0 mm and +17 mm.

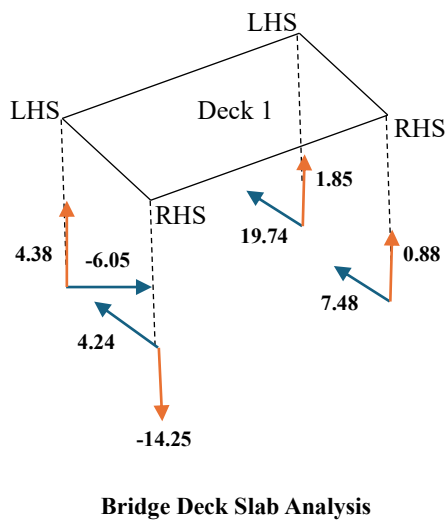
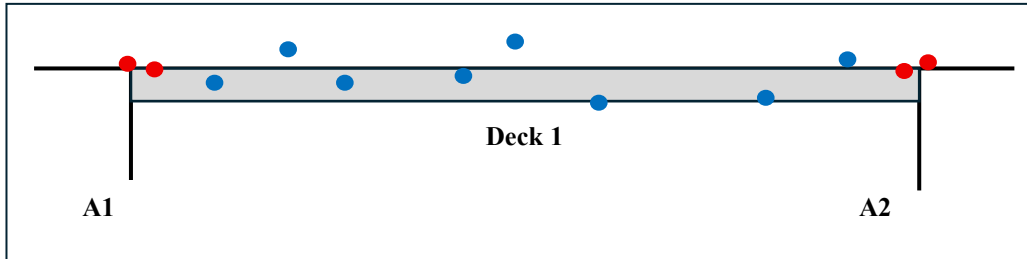
Critical Events & Anomalies:

- **April 2025:** VH rises to +14 mm, while VV transitions to –2 mm, indicating renewed lateral intensification accompanied by vertical settlement.
- **June 2025:** VH peaks at +17 mm, while VV remains slightly negative (–5 mm), confirming continued lateral displacement growth toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits pronounced non-uniform vertical displacement, indicating strong differential vertical response between the LHS and RHS regions. The coexistence of uplift on both sides and significant settlement at the central support confirms vertical rotation combined with localized bending, rather than uniform vertical deflection

Horizontal Displacement Behaviour:

- Horizontal displacement shows very strong directional variability, clearly indicating torsional deformation of the deck slab. The opposing horizontal movements across the deck strongly indicate torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by large lateral displacement combined with asymmetric vertical uplift and settlement. Overall, the magnitude and distribution of displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and support performance